

Practical 8

Aim : Use pytest to test a Python application

pythondemo.py

```
def add(a, b):  
    return a + b  
  
def subtract(a, b):  
    return a - b  
  
def multiply(a, b):  
    return a * b  
  
def divide(a, b):  
    if b == 0:  
        raise ValueError("Cannot divide by zero")  
    return a / b  
  
def is_even(num):  
    return num % 2 == 0
```

test_pythondemo.py

```
import pytest  
import pythondemo  
# Passing test  
def test_add():  
    assert pythondemo.add(2, 3) == 5  
# Failing test  
def test_multiply_fail():  
    assert pythondemo.multiply(4, 3) == 11  
# Exception test  
def test_divide_by_zero():  
    with pytest.raises(ValueError):  
        pythondemo.divide(10, 0)  
  
# Logical test  
def test_is_even():  
    assert pythondemo.is_even(4) is True  
    assert pythondemo.is_even(5) is False  
  
# Parametrized test  
@pytest.mark.parametrize("a,b,result", [  
    (1,2,3),  
    (2,2,4),  
    (5,5,11) # failing case  
)  
def test_add_param(a,b,result):  
    assert pythondemo.add(a,b) == result
```

Screenshot :

```
PS M:\Academics\Academics VI\OST\LAB\Practical 8\PythonDemo> python -m pytest -v
===== test session starts =====
platform win32 -- Python 3.14.3, pytest-9.0.2, pluggy-1.6.0 -- C:\Users\manth\AppData\Local\Python\pythoncore-3.14-64\python.exe
cachedir: .pytest_cache
rootdir: M:\Academics\Academics VI\OST\LAB\Practical 8\PythonDemo
collected 7 items

test_pythondemo.py::test_add PASSED [ 14%]
test_pythondemo.py::test_multiply_fail FAILED [ 28%]
test_pythondemo.py::test_divide_by_zero PASSED [ 42%]
test_pythondemo.py::test_is_even PASSED [ 57%]
test_pythondemo.py::test_add_param[1-2-3] PASSED [ 71%]
test_pythondemo.py::test_add_param[2-2-4] PASSED [ 85%]
test_pythondemo.py::test_add_param[5-5-11] FAILED [100%]

===== FAILURES =====
_____ test_multiply_fail _____

def test_multiply_fail():
> assert pythondemo.multiply(4, 3) == 11
E       assert 12 == 11
E       + where 12 = <function multiply at 0x0000019E166821F0>(4, 3)
E       +   where <function multiply at 0x0000019E166821F0> = pythondemo.multiply

test_pythondemo.py:8: AssertionError
_____ test_add_param[5-5-11] _____

a = 5, b = 5, result = 11

@pytest.mark.parametrize("a,b,result", [
```

